Q1:

Q17: How to reduce pattern count without affecting coverage?

Ans:- ① Tool itself will write optimized set of pattern. It will not write redundant pattern in the final pattern list.

② command :- order_patterns (given in ATPG)
- Reorders the internal pattern set.

- Test patterns can be re-arranged to detect most faults first, which is useful for:

(i) Truncating test pattern set in order to fit in ATE memory

(ii) Detecting failing chips as early as possible during testing.

This can actually be done so that if we first place the pattern which detects most of the faults, and if suppose the pattern set is very large then some of the patterns from end can be truncated and it will not cause much problem.

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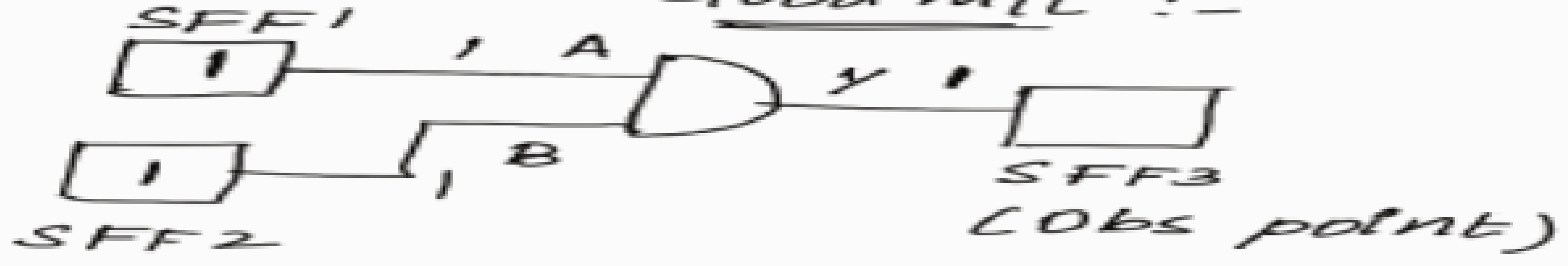
Q2 :

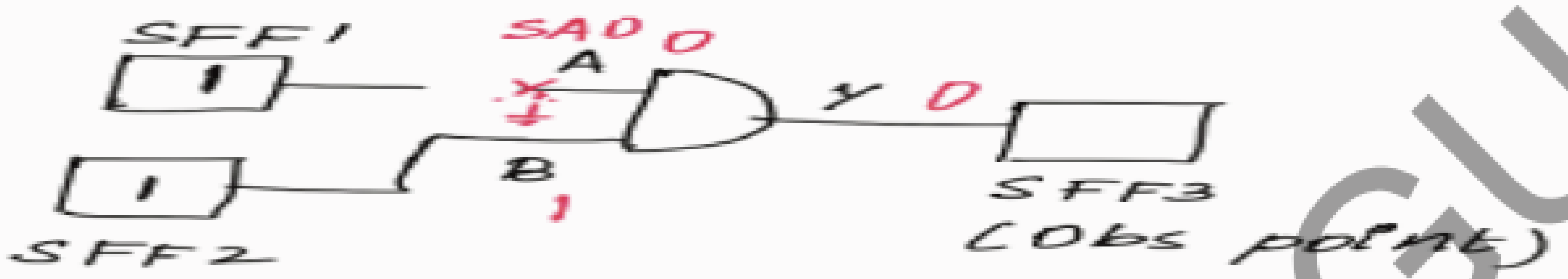
Q18: Explain SA fault by using waveform.

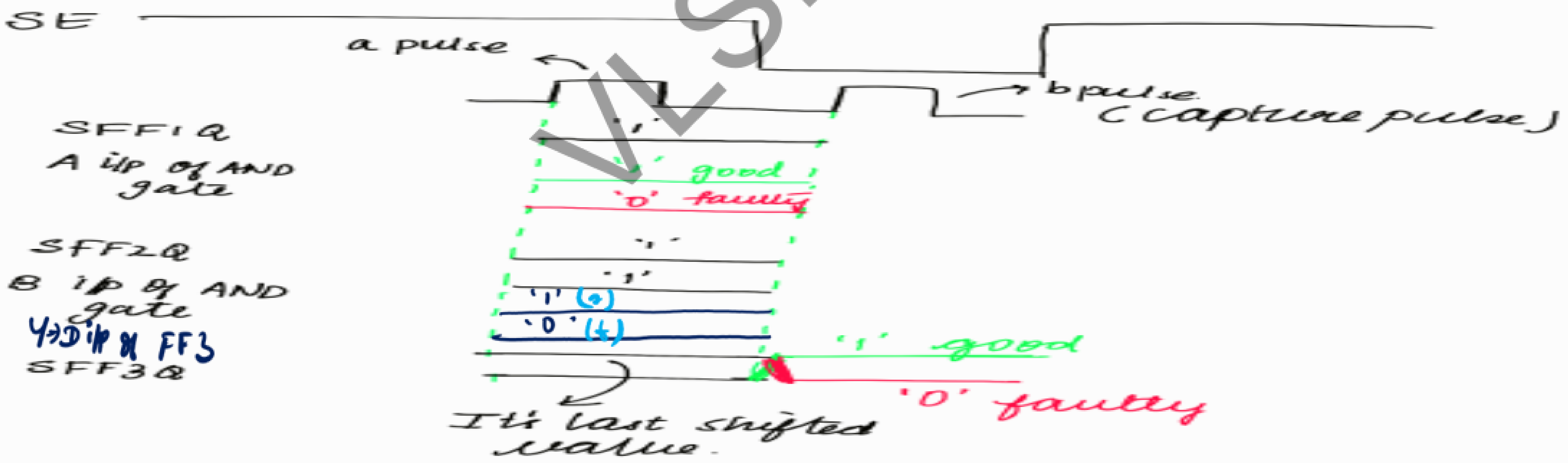
Ans :- Eg :- 2 i/p AND gate (A SA0)

Good m/c :-



Faulty m/c :-





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Q3 :

Q17: If we provide controllability and Observability to PIs and POs, will it improve the test coverage and how much % of improvement can we see?

NOTE :- Usually we don't connect the PI and PO to the tester (ATE) so that we can test more no. of chips at a time (theory AU.MPD)

If we connect PI and PO to the tester, will it improve the test cov and how much %. improvement can we see?

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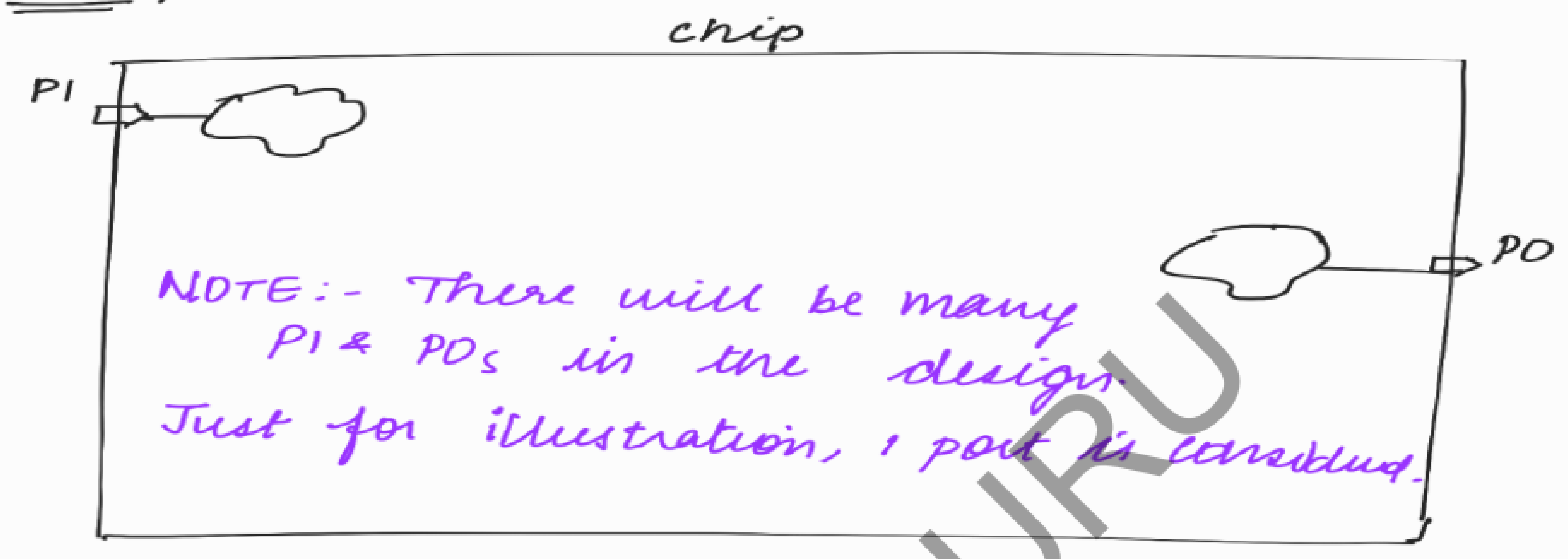
Ans :-



If we don't connect PI and PO to the tester, ~~then~~ so we can't test the logics that needs controllability from PI and the logics whose only obs point is PO as shown in fig.

we will mask the PO during ATPG

and we will constrain PI to X during ATPG.

So if we connect the PI and PO to the tester (provide controllability & observability to PI and PO), we will be able to test those logics.

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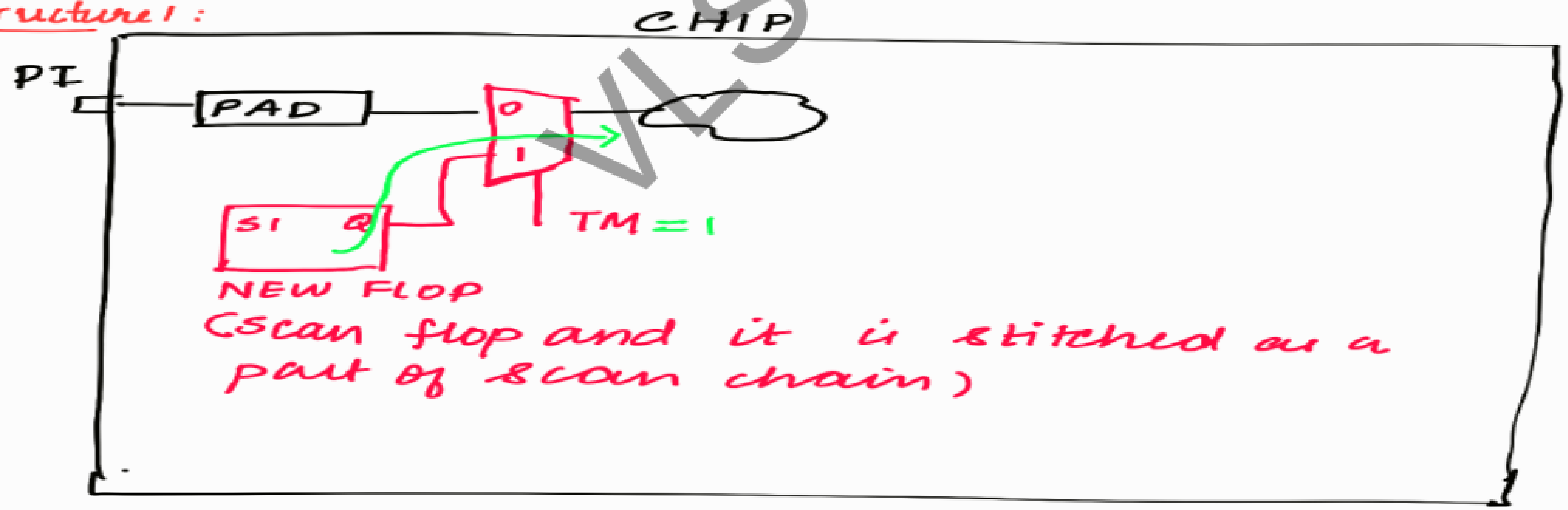
So our test cov will improve.
The % of improvement will depend on what portion of logics are controlled from PI and what portion of logic has only PO as its obs point.

NOTE :-
In real time projects, we usually add the below 2 structures for the logics which is controlled from PI and whose only observation point is PO.

Structure 1 :



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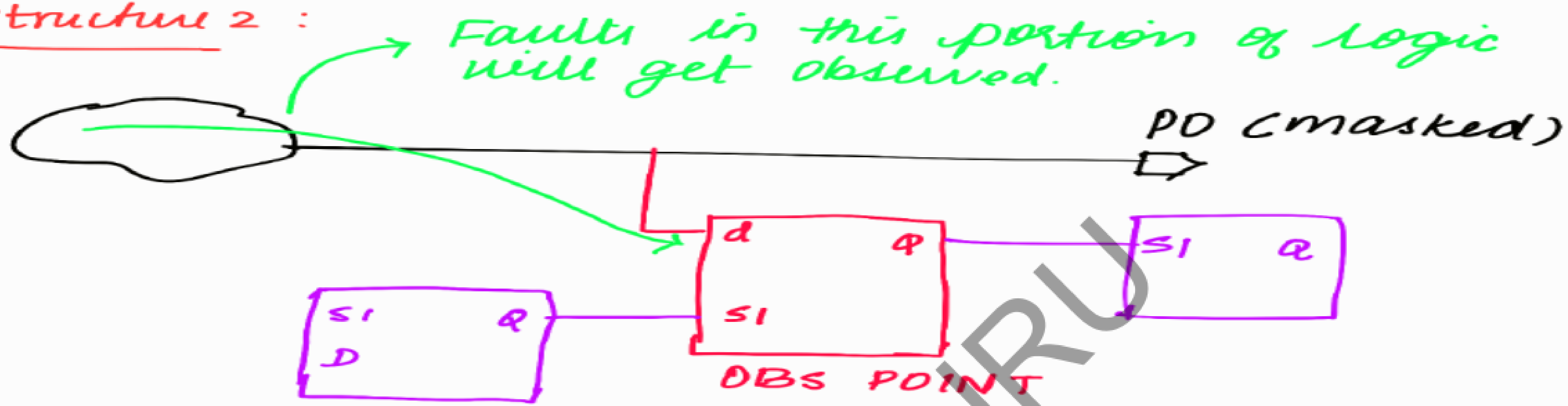
Structure 2 :



So in companies, by using the above 2 structures we have overcome the cov drop caused due to uncontrollable PI and unobservable PO.

We have overcome the problem of not connecting the PI, PO to the tester by adding the above 2 logic.

So we usually don't connect PI and PO to the tester so that we can test more no. of chips at a time.

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Q4:

Q20: While doing TDF, eventhough it is just 2 fast freq clock pulses in capture, it should still damage the chip.
How is it taken care?

Ans:- ① In original company projects, for examp
if there are 2 fn domains, it will be tested in 2 separate pattern lists.

Eg:- there are 2 fn clks:- clka and clkb
One TDF ATPG run will be done by gating clkb and allowing only clka.

Another TDF ATPG run will be done by gating clka and allowing only clkb.

So different pattern sets are generated.

If all ffs are toggled at the same time, it will cause power issues in silicon & hence it will damage the chip.

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But as per the above explanation, for 1 pattern set, we are allowing only 1 clock (Eg:- pattern set 1, we are allowing only clka)

∴ only those ffs whose clock is clka will toggle for that pattern set.

② power grid will be designed to handle this.

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Q5:

Q: Test plan for ATPG.

Ans :- We can decide the following :

(i) Expected cov values

(ii) Any target for test time based on previous project.

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Q6:

Q15: Reset is considered as clock in order to get coverage during stuck-at pattern generation?

But in atspeed pattern generation, reset is not covered. Why?

Ans:- Usually in designs we don't time these types of paths.

When reset gets de-asserted, either clocks have to be off or large-gap b/w 2 pulses is required inorder to avoid metastable condition. (Removal & recovery timing check → STA concept).

In TDF capture phase, the 2 pulses come at high freq. If you try to cover TDF faults, then the system will go into metastable state.

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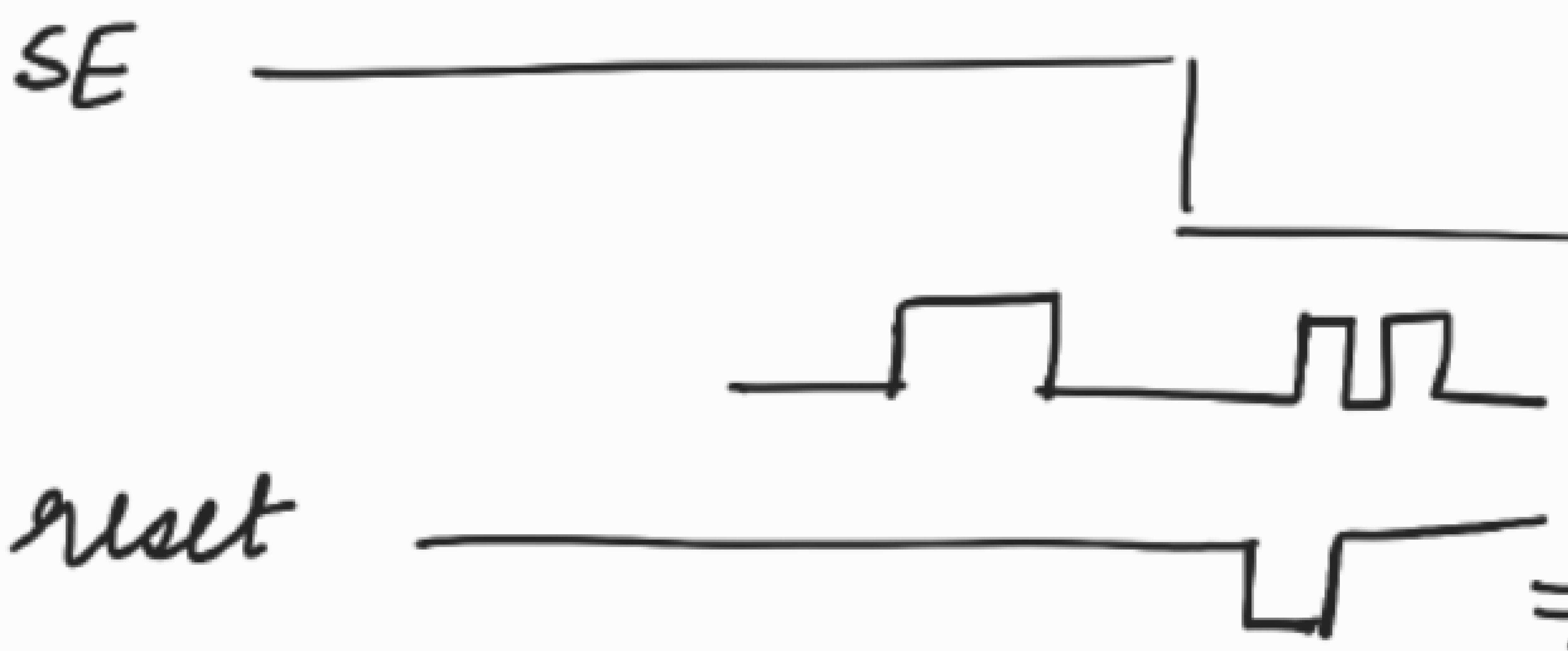

⇒ result in metastability.

NOTE:-

Not covering TDF for reset is not a problem as resets are not toggled at a high rate in designs.

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Q7:

Q2: Why do we use slow freq clock in the capture phase for stuck at and fast freq clock in the capture phase for TDF?

Ans :- For SA, we are seeing whether a net is stuck at 0 or 1.

That can be checked with slow freq clock itself.

But in TDF, we are checking whether the transition of signals happens within the time period of fast freq clock or not.

So we should use fast freq clock in the capture phase for TDF.

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